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HARNESS ISOLATION PLATE ASSEMBLY MECHANISM AND BATTERY PRODUCTION LINE

Abstract

A harness isolation plate assembly mechanism and a battery production line are disclosed. The harness isolation plate assembly mechanism includes: a suction tool including a first bracket and a suction assembly; where the first bracket is provided with a first connection structure being configured to connect to a transport device to move the first bracket, and an extension surface of the first bracket being parallel to an extension surface of a harness isolation plate to be grabbed; and the suction assembly is fixed on the extension surface of the first bracket and arranged to avoid a harness region of the harness isolation plate, and the suction assembly is configured to adsorb or release the harness isolation plate. The harness isolation plate assembly mechanism can avoid low-strength regions on the harness isolation plate and disperse a force on the harness isolation plate, achieving damage-free assembly of the harness isolation plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International application PCT/CN2024/110088 filed on Aug. 6, 2024 that claims priority to Chinese Patent Application No. 202322657126.3, filed on Sep. 28, 2023. The content of these applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to the field of battery technologies, and particularly to a harness isolation plate assembly mechanism and a battery production line.

BACKGROUND

[0003] The application of new energy batteries in life and industry is becoming increasingly widespread. For example, new energy vehicles equipped with batteries have been widely used, and batteries are also increasingly being used in the field of energy storage and other fields.

[0004] In a traditional traction battery pack, a harness busbar is typically provided to connect battery cell groups in series and parallel, each harness in the harness busbar reaches an electrode, and temperature sensors and voltage sampling harnesses are arranged on the surface of the harness busbar. This arrangement not only results in complex wiring but also occupies a significant amount of space.

[0005] To replace the original complex wiring, integrated processing has been applied to traction battery packs in the prior art. After integration, battery cell groups within the battery pack are connected in series and parallel via a harness isolation plate (Cells Contact System, CCS). The harness isolation plate mainly includes components such as palladium sheets, nickel sheets, thermistors, and harnesses. During the assembly process, a robotic arm grabs the harness isolation plate to assemble the harness isolation plate into the battery pack. However, due to the generally low strength of the components within the harness isolation plate, damage easily occurs during grabbing, significantly reducing the quality and efficiency of assembly. Therefore, reducing damage to the components within the harness isolation plate during the assembly process is a critical issue that urgently needs to be addressed.

SUMMARY

[0006] In view of the preceding technical problems, this disclosure provides a harness isolation plate assembly mechanism and a battery production line, thereby achieving damage-free assembly of the harness isolation plate.

[0007] According to a first aspect, this disclosure provides a harness isolation plate assembly mechanism including: a suction tool including a first bracket and a suction assembly; where the first bracket is provided with a first connection structure, the first connection structure being configured to connect to a transport device to move the first bracket, and an extension surface of the first bracket being parallel to an extension surface of a harness isolation plate to be grabbed; and the suction assembly is fixed on the extension surface of the first bracket and is arranged to

avoid a harness region of the harness isolation plate, and the suction assembly is configured to adsorb or release the harness isolation plate.

[0008] Since the first bracket is provided with the first connection structure connected to the transport device, the transport device can drive the suction tool to move via the first connection structure. The first bracket is provided with the suction assembly avoiding the harness region, and when the suction assembly adsorbs the harness isolation plate, the suction assembly avoids the harness region with lowest strength and only becomes in contact with non-harness regions with higher strength on the harness isolation plate, thereby avoiding damage to the harness. Additionally, compared to the traditional robotic arm grabbing the harness isolation plate, using the suction assembly to adsorb the harness isolation plate allows the force required to lift the harness isolation plate to be distributed across the extension surface of the harness isolation plate, avoiding damage to the harness isolation plate due to concentrated force.

[0009] In some embodiments, the suction assembly is arranged at a position on the first bracket away from an edge and corresponds to a palladium sheet of the harness isolation plate.

[0010] By arranging the suction assembly at the position on the first bracket away from the edge, the suction assembly is close to the center of the first bracket. By arranging the harness isolation plate corresponding to the palladium sheet, the suction assembly only becomes in contact with the palladium sheet during adsorption, so that a force-bearing point is concentrated only on the palladium sheet, which has the highest strength among the components of the harness isolation plate, avoiding components with lower strength on the harness isolation plate and preventing damage to the low-strength components due to the adsorption force.

[0011] In some embodiments, the suction assembly includes a suction cup and a suction cup base, the suction cup base being connected to the first bracket, and the suction cup being arranged on the suction cup base and configured to adsorb or release the harness isolation plate.

[0012] The suction cup is arranged on the first bracket via the suction cup base, the suction cup is arranged to avoid the harness region of the harness isolation plate, and the suction cup can adsorb or release the harness isolation plate. During adsorption, the suction cup only becomes in contact with the non-harness region with higher strength, thereby avoiding damage to the harness region with the lowest strength.

[0013] In some embodiments, the first bracket is provided with multiple groups of first fixing structures, and the suction cup base is detachably connected to the first bracket via any one group of the first fixing structures.

[0014] Since the suction cup base is detachably connected to the first bracket via the first fixing structure, the suction cup base is provided with the suction cup. The suction cup is frequently in contact with the harness isolation plate, resulting in wear. Under the condition that the suction cup is damaged, the suction cup base can be removed from the first fixing structure and replaced with a new suction cup, avoiding delays in assembly progress due to suction cup damage. Moreover, since the first bracket is provided with multiple groups of first fixing structures, the suction cup base can be detached from one first fixing structure and installed on another first fixing structure, thereby changing an adsorption position of the suction cup, enabling the suction cup to avoid harness regions of various models of harness isolation plates, enhancing compatibility of the suction tool, and achieving rapid type switching.

[0015] In some embodiments, the first fixing structure are multiple groups of suction cup base mounting holes arranged at different positions on the first bracket.

[0016] Since the first fixing structure includes the suction cup base mounting holes, the suction cup base cooperates with the suction cup base mounting holes so as to achieve detachment and installation, allowing convenient and rapid insertion and extraction, and enabling quick adjustment of the suction cup position.

[0017] In some embodiments, the suction cup base is movably connected to the first bracket, so that the suction cup base is movable relative to the first bracket to at least two different positions.

[0018] Since the suction cup base is movably connected to the first bracket, the suction cup base is provided with the suction cup, and the suction cup base moves relative to the first bracket, thereby changing the position of the suction cup. Under the condition that a model of the harness isolation plate to be adsorbed changes, the suction cup base can be moved to enable the suction cup to avoid harness regions of different types of harness isolation plates, enhancing the compatibility of the suction tool.

[0019] In some embodiments, the suction tool further includes a movable mechanism, the movable mechanism being connected to the suction cup base and configured to drive the suction cup base to be movable relative to the first bracket.

[0020] The suction cup base is movable relative to the first bracket via the movable mechanism, and the suction cup base is provided with the suction cup, thereby driving the suction cup to move relative to the first bracket. Under the condition that the model of the harness isolation plate to be adsorbed changes, the suction cup base can be moved to enable the suction cup to avoid harness regions of different types of harness isolation plates, enhancing the compatibility of the suction tool.

[0021] In some embodiments, the movable mechanism includes a slide rail arranged on the first bracket and a cylinder capable of driving the suction cup base to move, the suction cup base being slidably connected to the slide rail.

[0022] By arranging the slide rail on the first bracket, the suction cup base is slidably connected to the slide rail, achieving movement of the suction cup base relative to the first bracket. Simply pushing the suction cup base can change the position of the suction cup, making the suction tool compatible with different types of harness isolation plates. The operation is simple, type switching is fast, and time and effort are saved. By providing the cylinder capable of driving the suction cup base, electrical operation can be achieved, offering convenience and avoiding errors from manual operation.

[0023] In some embodiments, the suction tool further includes a first positioning structure arranged on the first bracket, the first positioning structure having a positioning portion arranged along a suction direction of the suction assembly, and the first positioning structure being configured to match a corresponding structure of another tool to prevent relative displacement between the suction tool and the another tool along a non-suction direction.

[0024] By arranging the first positioning structure on the first bracket, and since the first positioning structure can match the corresponding structure of the other tool, positioning between the suction tool and the other tool is achieved. Since the first positioning structure has the positioning portion arranged along the suction direction of the suction assembly, the suction tool and the another tool cannot displace along the non-suction direction. The structure is simple, and assembly precision is high.

[0025] In some embodiments, the first positioning structure includes a positioning pin oriented toward the suction direction of the suction assembly, the positioning pin being configured to cooperate with a corresponding structure of an assembly platform to prevent relative displacement between the suction tool and the assembly platform along the non-suction direction.

[0026] Since the positioning pin cooperates with the corresponding structure of the assembly platform, and the positioning pin is oriented toward the suction direction of the suction assembly, the suction tool and the assembly platform can be positioned via the positioning pin, preventing displacement between the suction tool and the assembly platform along the non-suction direction.

[0027] In some embodiments, the first positioning structure includes a positioning groove arranged on the first bracket, the positioning groove being configured to cooperate with a corresponding structure of a positioning platform to prevent relative displacement between the suction tool and the positioning platform along the non-suction direction.

[0028] Since the first bracket is provided with the positioning groove, the positioning groove can cooperate with the corresponding structure of the positioning platform, and after the corresponding

structure of the positioning platform cooperates with the positioning groove, displacement between the suction tool and the positioning platform along the non-suction direction is prevented.

[0029] In some embodiments, the suction tool further includes a buffer member, the buffer member being arranged on a side of the suction assembly facing the harness isolation plate, and under the condition that the suction assembly adsorbs the harness isolation plate, the buffer member abutting against the harness isolation plate.

[0030] Since the buffer member is provided on the side of the suction assembly facing the harness isolation plate, when the suction assembly adsorbs the harness isolation plate, the harness isolation plate directly becomes in contact with the buffer member rather than the suction assembly, and the buffer member disperses and cushions pressure between the harness isolation plate and the suction assembly.

[0031] In some embodiments, the buffer member is an elastic pad arranged on the suction assembly, the elastic pad is provided with multiple suction holes, and the suction assembly adsorbs the harness isolation plate through the suction holes.

[0032] Since the elastic pad is provided on the side of the suction assembly facing the harness isolation plate, during adsorption, the elastic pad directly becomes in contact with the harness isolation plate, dispersing and cushioning the pressure between the harness isolation plate and the suction assembly through the elasticity of the elastic pad. The elastic pad is provided with the suction holes, enabling the suction assembly to perform normal adsorption through the suction holes.

[0033] In some embodiments, the suction tool further includes a vacuum generator assembly arranged on the first bracket, the vacuum generator assembly being in communication with the suction assembly and configured to provide power to the suction assembly for adsorbing the harness isolation plate.

[0034] By arranging the vacuum generator assembly on the first bracket, power required for the suction assembly to adsorb the harness isolation plate is provided.

[0035] In some embodiments, the vacuum generator assembly includes a vacuum generator, a pneumatic switch valve, and a pneumatic duplex component, the vacuum generator being sequentially connected to the suction assembly via the pneumatic duplex component and the pneumatic switch valve.

[0036] Power for the suction assembly to adsorb the harness isolation plate is provided through the vacuum generator. The pneumatic duplex component controls an opening degree of the pneumatic switch valve based on the air pressure within the suction assembly, thereby stabilizing the air pressure within the suction assembly. The pneumatic duplex component is further configured to filter air, filtering air from the suction assembly.

[0037] In some embodiments, a safety hole is provided on the first bracket at a position corresponding to the suction assembly, and the safety hole is in communication with the suction assembly.

[0038] By providing the safety hole in communication with the suction assembly on the first bracket, air circulation is maintained, preventing excessively low air pressure inside the suction assembly.

[0039] In some embodiments, the first connection structure is a first lifting ring arranged on the first bracket.

[0040] Connection of the first lifting ring and the transport device allows the suction tool to be hoisted to a designated position.

[0041] In some embodiments, the harness isolation plate assembly mechanism further includes a positioning tool arranged on a positioning platform, the positioning tool includes a second bracket and a second positioning structure, the second bracket is configured to support the harness isolation plate; and the second positioning structure is arranged on the second bracket and configured to cooperate with the first positioning structure to prevent relative displacement between the suction

tool and the positioning platform along the non-suction direction.

[0042] Since the positioning tool includes the second positioning structure, and the second positioning structure can cooperate with the first positioning structure, enabling mutual positioning between the positioning tool and the suction tool. Since the first positioning structure has the positioning portion arranged along the suction direction of the suction assembly, relative displacement between the suction tool and the positioning platform along the non-suction direction can be prevented.

[0043] In some embodiments, the second positioning structure is a top post arranged on the second bracket, the top post protrudes upward along the second bracket, and under the condition that the top post cooperates with the first positioning structure, the top post supports and lifts the suction tool, so that the suction assembly and an upper surface of the second bracket have a clearance gap.

[0044] Since the second bracket is provided with the top post protruding upward, after the top post cooperates with the first positioning structure, displacement between the positioning tool and the suction tool in the non-suction direction can be prevented, and the suction tool can be lifted by the top post, maintaining a safety gap between the positioning tool and the suction tool.

[0045] In some embodiments, the positioning tool further includes a limiting member arranged on the second bracket, the limiting member being configured to enclose a positioning space matching an outer contour of the harness isolation plate.

[0046] By arranging the limiting member on the second bracket, and since the limiting member encloses the positioning space matching the outer contour of the harness isolation plate, after the suction tool moves the harness isolation plate to the positioning tool and positioning is achieved through cooperation between the first positioning structure and the second positioning structure, the harness isolation plate is released so that the harness isolation plate to fall into the positioning space enclosed by the limiting member, achieving position correction of the harness isolation plate.

[0047] In some embodiments, a side of the limiting member facing the positioning space is provided with a guide slope, and an upper end of the guide slope is inclined toward a direction away from the positioning space, so that a planar contour of an end of the positioning space away from the second bracket is larger than a planar contour of an end of the positioning space close to the second bracket.

[0048] Since the guide slope is provided on the side of the limiting member facing the positioning space, and the upper end of the guide slope is inclined toward the direction away from the positioning space, even if the falling harness isolation plate has a certain deviation relative to the positioning space, the harness isolation plate can be easily guided into the positioning space.

[0049] In some embodiments, the limiting member is slidably connected to the second bracket, allowing the limiting member to slide and adjust a size of the positioning space.

[0050] By slidably connecting the limiting member to the second bracket, the limiting member can slide relative to the second bracket to change a position of the limiting member relative to the second bracket, thereby changing a distance between the limiting members and adjusting the size of the positioning space. This allows the limiting member to be suitable for positioning harness isolation plates of different sizes, offering strong compatibility.

[0051] In some embodiments, the second bracket is provided with multiple support members, the multiple support members are arranged to avoid the harness region of the harness isolation plate to support the harness isolation plate.

[0052] By arranging the support members avoiding the harness region on the second bracket, after the harness isolation plate is released and falls onto the positioning tool, the harness region with the lowest strength on the harness isolation plate is suspended, and only non-harness regions are in contact with the support members. This prevents impact force from damaging the harness region when the harness isolation plate falls onto the positioning tool.

[0053] In some embodiments, the second bracket is further provided with a second connection structure, the second connection structure being configured to connect to the transport device to

move the positioning tool.

[0054] The transport device can drive the positioning tool to move via the second connection structure.

[0055] In some embodiments, the positioning tool further includes support legs, the support legs being connected to the second bracket to support the second bracket and adjust a height of the second bracket.

[0056] By arranging the support legs on the second bracket, support for the positioning tool is achieved. Additionally, the support legs can be further configured to adjust the height of the second bracket, facilitating height adaptation between the positioning tool and the suction tool.

[0057] In some embodiments, the harness isolation plate assembly mechanism further includes a loading tray, the loading tray being arranged on a loading platform and configured to support the harness isolation plate to be adsorbed; and the loading tray being provided with a third positioning structure configured to cooperate with the first positioning structure to prevent relative displacement between the suction tool and the loading platform along the non-suction direction.

[0058] Since the loading tray is provided with the third positioning structure cooperating with the first positioning structure, and the first positioning structure is arranged on the suction tool, after the first positioning structure cooperates with the third positioning structure, a position between the suction tool and the loading tray is fixed. Since the first positioning structure has the positioning portion arranged along the suction direction of the suction assembly, relative displacement between the suction tool and the loading platform along the non-suction direction can be prevented.

[0059] According to a second aspect, this disclosure provides a battery production line including: an assembly platform; the harness isolation plate assembly mechanism as described above; and a transport device, the transport device being at least configured to transport the harness isolation plate assembly mechanism to the assembly platform.

[0060] Since the battery production line is equipped with the harness isolation plate assembly mechanism described above, during adsorption of the harness isolation plate, low-strength regions on the harness isolation plate can be avoided, and the force on the harness isolation plate is dispersed, achieving damage-free assembly of the harness isolation plate.

[0061] In some embodiments, the harness isolation plate assembly mechanism includes a suction tool, a positioning tool, and a loading tray; and the battery production line further includes a loading platform and a positioning platform, the loading tray being arranged on the loading platform, the positioning tool being arranged on the positioning platform; and the transport device being at least configured to transport the suction tool from the loading platform to the positioning platform and from the positioning platform to the assembly platform.

[0062] The suction tool adsorbs the harness isolation plate from the loading tray on the loading platform, the transport device moves the suction tool to the positioning platform, the suction tool releases the harness isolation plate to the positioning tool on the positioning platform for positioning, and the suction tool adsorbs the positioned harness isolation plate to the assembly platform to complete subsequent assembly. This achieves precise assembly.

[0063] In some embodiments, the suction tool is provided with a first positioning structure, the positioning tool is provided with a second positioning structure, the loading tray is provided with a third positioning structure, and the assembly platform is provided with a fourth positioning structure; where the first positioning structure is capable of cooperating with the second positioning structure to restrict relative displacement between the suction tool and the positioning tool; the first positioning structure is capable of cooperating with the third positioning structure to restrict relative displacement between the suction tool and the loading tray; and the first positioning structure is capable of cooperating with the fourth positioning structure to restrict relative displacement between the suction tool and the assembly platform.

[0064] Positioning between the suction tool and the positioning tool can be achieved through cooperation between the first positioning structure and the second positioning structure; positioning

between the suction tool and the loading tray can be achieved through cooperation between the first positioning structure and the third positioning structure; and positioning between the suction tool and the assembly platform can be achieved through cooperation between the first positioning structure and the fourth positioning structure. This achieves precise assembly.

[0065] The foregoing description is merely an overview of the technical solution of this disclosure. For a better understanding of the technical means in this disclosure so that they can be implemented according to the content of the specification, and to make the above and other objectives, features, and advantages of this disclosure more obvious and easier to understand, the following describes specific embodiments of this disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0066] Persons of ordinary skill in the art can clearly understand various other advantages and benefits by reading the detailed description of the preferred embodiments below. The accompanying drawings are merely intended to illustrate the preferred implementations and are not intended to limit this disclosure. Moreover, throughout the accompanying drawings, the same reference signs represent the same parts. In the accompanying drawings:

[0067] FIG. 1 is a schematic diagram of a harness isolation plate according to some embodiments of this disclosure;

[0068] FIG. 2 is a three dimensional diagram of a suction tool of a harness isolation plate according to some embodiments of this disclosure;

[0069] FIG. 3 is a side view of a suction tool of a harness isolation plate according to some embodiments of this disclosure;

[0070] FIG. 4 is a bottom view of a suction tool of a harness isolation plate according to some embodiments of this disclosure;

[0071] FIG. 5 is a three dimensional diagram of a positioning tool for a harness isolation plate according to some embodiments of this disclosure;

[0072] FIG. 6 is a side view of a positioning tool of a harness isolation plate according to some embodiments of this disclosure;

[0073] FIG. 7 is a schematic diagram of cooperation between a suction tool and a positioning tool according to some embodiments of this disclosure; and

[0074] FIG. 8 is a schematic diagram of a suction tool operating on an assembly platform according to some embodiments of this disclosure.

REFERENCE SIGNS ARE DESCRIBED AS FOLLOWS

[0075] **10.** harness isolation plate; **101.** harness; **102.** palladium sheet; **103.** plastic plate; **12.** suction tool; **121.** first bracket; **122.** suction assembly; **1221.** suction cup; **1222.** suction cup base mounting hole; **123.** first connection structure; **1231.** first lifting ring; **1232.** handle; **124.** positioning pin; **125.** buffer member; **1251.** suction hole; **126.** vacuum generator; **127.** pneumatic switch valve; **128.** pneumatic duplex component; **129.** safety hole; **130.** positioning groove; **14.** positioning tool; **141.** second bracket; **142.** second positioning structure; **143.** limiting member; **144.** oblong hole; **145.** support member; **146.** second connection structure; **147.** support leg; **16.** assembly platform; and **161.** fourth positioning structure.

DETAILED DESCRIPTION

[0076] The following will describe the embodiments of the technical solutions of this disclosure in detail with reference to the accompanying drawings. The following embodiments are only for a clearer explanation of the technical solutions of this disclosure, and therefore they are only examples and cannot be used to limit the scope of protection of this disclosure.

[0077] Unless otherwise defined, all technical and scientific terms used herein have the same

meanings as commonly understood by persons of ordinary skill in the art to which this disclosure belongs; the terms used herein are only for the purpose of describing specific embodiments and are not intended to limit this disclosure; the terms “include” and “have” and any variations thereof in this disclosure are intended to cover non-exclusive inclusions.

[0078] In the description of the embodiments of this disclosure, the technical terms “first”, “second”, “third”, and the like are only used to distinguish different objects and cannot be understood as indicating or implying relative importance or implicitly indicating the number, specific order, or primary-secondary relationship of the indicated technical features. In the description of the embodiments of this disclosure, “multiple” means more than two unless otherwise explicitly specified.

[0079] As mentioned herein, “embodiment” means that specific features, structures, or characteristics described with reference to an embodiment can be included in at least one embodiment of this disclosure. The word “embodiment” appearing in various places in this specification does not necessarily refer to the same embodiment or an independent or alternative embodiment that is exclusive of other embodiments. Persons skilled in the art explicitly and implicitly understand that the embodiments described herein may be combined with other embodiments.

[0080] In the description of the embodiments of this disclosure, the term “and/or” is only an associative relationship for describing associated objects, indicating that three relationships may be present. For example, A and/or B may indicate the following three cases: presence of only A, presence of both A and B, and presence of only B. In addition, a character “/” in this specification generally indicates an “or” relationship between contextually associated objects.

[0081] In the description of the embodiments of this disclosure, the technical terms “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “circumferential”, and the like indicating orientation or positional relationships are based on the orientation or positional relationships shown in the drawings, and are only for the convenience of describing the embodiments of this disclosure and simplifying the description, rather than indicating or implying that the apparatuses or elements referred to must have specific orientations, be constructed, operated, or used in specific orientations, and therefore should not be understood as limiting the embodiments of this disclosure.

[0082] In the description of the embodiments of this disclosure, unless otherwise specified and defined explicitly, the technical terms “mounting”, “connection”, “joining”, and “fastening” should be understood in their general senses. For example, they may refer to a fixed connection, a detachable connection, or an integral connection, may refer to a mechanical connection or an electrical connection, and may refer to a direct connection, an indirect connection via an intermediate medium, an internal communication between two elements, or an interaction between two elements. Persons of ordinary skill in the art can understand specific meanings of these terms in this disclosure as appropriate to specific situations.

[0083] In the description of the embodiments of this disclosure, unless otherwise specified and defined explicitly, the technical term “contact” should be broadly understood, and may be direct contact or contact through an intermediate medium layer, may be contact between two objects with essentially no interaction force, or may be contact between two objects with interaction force.

[0084] This disclosure will be described in detail below.

[0085] Currently, the application of new energy batteries in life and industry is becoming increasingly widespread. New energy batteries have been widely used in energy storage power supply systems such as hydroelectric power plants, thermal power plants, wind power plants, and solar power plants, as well as in many other fields including electric transportation tools such as electric bicycles, electric motorcycles, and electric vehicles, and aerospace.

[0086] In the embodiments of this disclosure, a battery may include one or more battery cells to provide voltage and capacity. When there are multiple battery cells, the multiple battery cells are

connected in series, parallel, or series-parallel through busbar.

[0087] In some embodiments, the battery may be a battery pack (battery Pack), the battery pack includes a box and one or more battery cells, and the battery cells are accommodated in the box.

[0088] In a traditional traction battery pack, a harness busbar is typically provided to connect battery cell groups in series and parallel, each harness in the harness busbar reaches an electrode, and temperature sensors and voltage sampling harnesses are arranged on the surface of the harness busbar. This arrangement not only results in complex wiring but also occupies a significant amount of space.

[0089] To replace the original complex wiring, integrated processing has been applied to traction battery packs in the prior art. After integration, battery cell groups within the battery pack are connected in series and parallel via a harness isolation plate (Cells Contact System, CCS). The harness isolation plate mainly includes components such as palladium sheets, nickel sheets, thermistors, and harnesses. During the assembly process, a robotic arm grabs the harness isolation plate to assemble the harness isolation plate into the battery pack. However, due to the generally low strength of the components within the harness isolation plate, damage easily occurs during grabbing, significantly reducing the quality and efficiency of assembly. Therefore, reducing damage to the components within the harness isolation plate during the assembly process is a critical issue that urgently needs to be addressed.

[0090] The inventors of this disclosure have noted that when grabbing the harness isolation plate with a robotic arm, a clamping end of the robotic arm compresses components on the harness isolation plate, causing excessive local force on the harness isolation plate and resulting in damage to the harness isolation plate. Currently, most existing automated assembly mechanisms for harness isolation plates do not take into account for the issue that the harness isolation plates themselves have low strength, making it difficult to guarantee the quality of the assembled harness isolation plates.

[0091] In response to the deficiencies in the prior art mentioned above, the inventors of this disclosure have found that if low-strength regions on the harness isolation plate can be avoided and the force required to lift the harness isolation plate can be distributed on an extension surface of the harness isolation plate, the assembly of the harness isolation plate can be completed without damaging the harness isolation plate.

[0092] Based on this design concept, the inventors of this disclosure have designed a harness isolation plate assembly mechanism including: a suction tool including a first bracket and a suction assembly; where the first bracket is provided with a first connection structure, the first connection structure being configured to connect to a transport device to move the first bracket, and an extension surface of the first bracket being parallel to an extension surface of a harness isolation plate to be grabbed; and the suction assembly is fixed on the extension surface of the first bracket and is arranged to avoid a harness region of the harness isolation plate, and the suction assembly is configured to adsorb or release the harness isolation plate.

[0093] Since the first bracket is provided with the first connection structure connected to the transport device, the transport device can drive the suction tool to move via the first connection structure. The first bracket is provided with the suction assembly avoiding the harness region, and when the suction assembly adsorbs the harness isolation plate, the suction assembly avoids the harness region with lowest strength and only becomes in contact with non-harness regions with higher strength on the harness isolation plate, thereby avoiding damage to the harness. Additionally, compared to the traditional robotic arm grabbing the harness isolation plate, using the suction assembly to adsorb the harness isolation plate allows the force required to lift the harness isolation plate to be distributed across the extension surface of the harness isolation plate, avoiding damage to the harness isolation plate due to concentrated force.

[0094] The harness isolation plate assembly mechanism provided in the embodiments of this disclosure may be used for, but is not limited to, the harness isolation plate, and may also be used

for assembling other suitable workpieces, such as printed circuit boards and flexible circuit boards.

[0095] The harness isolation plate assembly mechanism provided in the embodiments of this disclosure may be used in production lines such as battery production lines, for adsorbing workpieces in intermediate stages of production and moving the workpieces to other locations.

[0096] The following description is provided with reference to the accompanying drawings.

[0097] FIG. 1 is a schematic diagram of a harness isolation plate according to some embodiments of this disclosure; FIG. 2 is a three dimensional diagram of a suction tool of a harness isolation plate according to some embodiments of this disclosure; FIG. 3 is a side view of a suction tool of a harness isolation plate according to some embodiments of this disclosure; FIG. 4 is a bottom view of a suction tool of a harness isolation plate according to some embodiments of this disclosure; FIG. 5 is a three dimensional diagram of a positioning tool of a harness isolation plate according to some embodiments of this disclosure; FIG. 6 is a side view of a positioning tool of a harness isolation plate according to some embodiments of this disclosure; FIG. 7 is a schematic diagram of cooperation between a suction tool and a positioning tool according to some embodiments of this disclosure; and FIG. 8 is a schematic diagram of a suction tool operating on an assembly platform according to some embodiments of this disclosure.

[0098] As shown in FIG. 2 to FIG. 4, a harness isolation plate **10** assembly mechanism includes: a suction tool **12** including a first bracket **121** and a suction assembly **122**; where the first bracket **121** is provided with a first connection structure **123**, the first connection structure **123** being configured to connect to a transport device to move the first bracket **121**, and an extension surface of the first bracket **121** being parallel to an extension surface of a harness isolation plate **10** to be grabbed; and the suction assembly **122** is fixed on the extension surface of the first bracket **121** and is arranged to avoid a harness region of the harness isolation plate **10**, and the suction assembly **122** is configured to adsorb or release the harness isolation plate **10**. The harness region is a region on the harness isolation plate **10** in which all harnesses **101** are located.

[0099] The harness isolation plate **10** assembly mechanism provided in this disclosure is applicable to the harness isolation plate **10**, printed circuit boards, flexible circuit boards, and the like. Taking the harness isolation plate as an example, as shown in FIG. 1, the harness isolation plate **10** consists of nickel sheets, palladium sheets **102**, thermistors, harnesses **101**, and a plastic plate **103**. The nickel sheets, the palladium sheets **102**, the thermistors, and the harnesses **101** are all arranged on the plastic plate **103**, with nickel sheets and palladium sheets **102** arranged on two sides of the harness **101** and each palladium sheet **102** connected to the harness **101**. Compared to other plates, the components constituting the harness isolation plate **10** are directly fixed on the plastic plate **103**, with no protective layers such as a hot-pressed protective film provided on a surface, meaning that the components on the harness isolation plate **10** are exposed and highly susceptible to damage. Among the components constituting the harness isolation plate **10**, the harnesses **101** have the lowest strength, while the palladium sheets **102** and the plastic plate **103** supporting the components have relatively high strength.

[0100] Continuing to refer to FIG. 2 to FIG. 4, the first bracket **121** may be configured as a plate shape, and a surface area of the first bracket **121** is not less than a surface area of the harness isolation plate **10**. The suction assembly **122** is arranged on one side surface of the first bracket **121**, and the first connection structure **123** is arranged on the other side surface of the first bracket **121**. The transport device is connected to the first connection structure **123** and drives the suction tool **12** to move via the first connection structure **123**. The first bracket **121** is configured as a straight plate, making the extension surface of the first bracket **121** a flat surface.

[0101] The suction assembly **122** is fixed on a surface of the first bracket **121** away from the first connection structure **123**, and the suction assembly corresponds to regions of higher strength on the harness isolation plate **10**, the high-strength regions including the plastic plate **103** and the palladium sheets **102**. The suction assembly avoids the harness region with lower strength on the harness isolation plate **10**. The suction assembly **122** may be evacuated to adsorb the harness

isolation plate **10**, or the suction assembly **122** may be turned off to release the harness isolation plate **10**. A surface of the suction assembly **122** being in contact with the harness isolation plate **10** is configured as a flat surface, and the flat surface is parallel to the extension surface of the first bracket **121**. When the suction assembly **122** adsorbs the harness isolation plate **10**, the extension surface of the first bracket **121** is parallel to the extension surface of the harness isolation plate **10**. [0102] Since the first bracket **121** is provided with the first connection structure **123** connected to the transport device, the transport device can drive the suction tool **12** to move via the first connection structure **123**. The first bracket **121** is provided with the suction assembly **122** avoiding the harness region, and when the suction assembly **122** adsorbs the harness isolation plate **10**, the suction assembly **122** avoids the harness region with lowest strength and only becomes in contact with non-harness regions with higher strength on the harness isolation plate **10**, thereby avoiding damage to the harness **101**. Additionally, compared to the traditional robotic arm grabbing the harness isolation plate **10**, using the suction assembly **122** to adsorb the harness isolation plate **10** allows the force required to lift the harness isolation plate **10** to be distributed across the extension surface of the harness isolation plate **10**, avoiding damage to the harness isolation plate **10** due to concentrated force.

[0103] In some embodiments, the suction assembly **122** corresponds to at least one of the palladium sheet **102** and the plastic plate **103** of the harness isolation plate **10**. Optionally, the suction assembly **122** is arranged at a position on the first bracket **121** away from an edge and corresponds to the palladium sheet **102** of the harness isolation plate **10**. An edge of the suction assembly **122** is away from an edge of the first bracket **121**, so that the suction assembly **122** is located at a position close to a center of the first bracket **121**. This makes the center of gravity of the suction tool **12** concentrated and close to the center. Therefore, during the process of the transport device transporting the suction tool **12**, the shape of the suction tool **12** can be kept stable, enabling the suction tool **12** to maintain a state parallel to the harness isolation plate **10**. When the suction assembly **122** adsorbs the harness isolation plate **10**, the suction assembly **122** can be distributed across a surface of the first bracket **121**, each suction assembly **122** corresponding to one palladium sheet **102** of the harness isolation plate **10**, so that an adsorption force provided by the suction assembly **122** is applied only to the palladium sheet **102** with higher strength.

[0104] By arranging the suction assembly **122** at the position on the first bracket **121** away from the edge, the suction assembly **122** is close to the center of the first bracket **121**. By arranging the harness isolation plate **10** corresponding to the palladium sheet **102**, the suction assembly **122** only becomes in contact with the palladium sheet **102** during adsorption, so that a force-bearing point is concentrated only on the palladium sheet **102**, which has the highest strength among the components of the harness isolation plate **10**, avoiding components with lower strength on the harness isolation plate **10** and preventing damage to the low-strength components due to the adsorption force.

[0105] In some embodiments, the suction assembly **122** includes a suction cup **1221** and a suction cup base, the suction cup base being connected to the first bracket **121**, and the suction cup **1221** being arranged on the suction cup base and configured to adsorb or release the harness isolation plate **10**. A pattern in which multiple suction cups **1221** are arranged on one suction cup base may be adopted, or a pattern in which the suction cup base is arranged in a one-to-one correspondence with the suction cup **1221** may be adopted. The suction cup **1221** is arranged on the suction cup base to avoid the harness region of the harness isolation plate **10** and is connected to the first bracket **121** via the suction cup base. When the suction tool **12** is used to adsorb the harness isolation plate **10**, the suction cup **1221** directly becomes in contact with the harness isolation plate **10**, airflow within the suction cup **1221** is used to adsorb the harness isolation plate **10**, and turning off the airflow within the suction cup **1221** releases the harness isolation plate **10**. Further, the suction cup **1221** corresponds to at least one of the palladium sheet **102** and the plastic plate **103** of the harness isolation plate **10**. Each suction cup **1221** may be arranged corresponding to one

palladium sheet **102**.

[0106] The suction cup **1221** is arranged on the first bracket **121** via the suction cup base, the suction cup **1221** is arranged to avoid the harness region of the harness isolation plate **10**, and the suction cup **1221** can adsorb or release the harness isolation plate **10**. During adsorption, the suction cup **1221** only becomes in contact with the non-harness region with higher strength, thereby avoiding damage to the harness region with the lowest strength.

[0107] In some embodiments, the first bracket **121** is provided with multiple groups of first fixing structures, and the suction cup base is detachably connected to the first bracket **121** via any one group of the first fixing structures. Since harness regions and dimensions vary among different models of harness isolation plates **10**, to enhance compatibility of the suction tool **12** with different models of wiring harness isolation boards **10**, multiple groups of suction assemblies **122** may be configured according to models of the harness isolation plate **10**, so that each group of suction assemblies **122** is suitable for different models of harness isolation plates **10**.

[0108] Specifically, corresponding to different types of harness isolation plates **10**, lengths of the suction cup bases are varied, and the multiple suction cups **1221** are arranged on each suction cup base. A length of each suction cup base matches a length of the harness isolation plate **10**. At least two suction cup bases form a group, each group of suction cup bases is mounted on the first bracket **121** via the first fixing structure, and after installation, the suction cup bases can avoid the harness region of the harness isolation plate **10**. Optionally, corresponding to different models of harness isolation plates **10**, the suction assembly **122** is configured such that the suction cup **1221** is arranged on the suction cup base avoiding the harness region, each group of suction assemblies **122** can avoid the harness region of one type of harness isolation plate **10**, and the suction assembly **122** is mounted on the first bracket **121** via the first fixing structure. Similarly, optionally, the suction assembly **122** is configured such that at least one suction cup **1221** is arranged on one suction cup base, and multiple groups of identical suction assemblies **122** are produced. Based on a dimension of the harness isolation plate **10**, a number of suction assemblies **122** mounted on the first bracket **121** is increased or decreased, so that a combination of the suction assemblies **122** corresponds to the dimension of the harness isolation plate **10**.

[0109] The first fixing structure is arranged on the first bracket **121** avoiding the harness region, and multiple groups of first fixing structures are provided based on positions of harness regions of different models of harness isolation plates **10**, so that each group of first fixing structures can respectively avoid the harness region of one model of harness isolation plate **10**. During use, a suction assembly **122** corresponding to a dimension of the harness isolation plate **10** is selected, which may be that a dimension of a suction assembly **122** corresponds to the dimension of the harness isolation plate **10**, or may be that a dimension of a combination of multiple groups of suction assemblies **122** corresponds to the dimension of the harness isolation plate **10**. Based on the harness region of the harness isolation plate **10**, a first fixing structure capable of avoiding the harness region is selected, and the suction assembly **122** is mounted on the first bracket **121** via this first fixing structure, so that a length of the suction assembly **122** is suitable for the harness isolation plate **10**, and a mounting position of the suction assembly **122** avoids the harness region of the harness isolation plate **10**.

[0110] Since the suction cup base is detachably connected to the first bracket **121** via the first fixing structure, the suction cup base is provided with the suction cup **1221**. The suction cup **1221** is frequently in contact with the harness isolation plate **10**, resulting in wear. Under the condition that the suction cup **1221** is damaged, the suction cup base can be removed from the first fixing structure and replaced with a new suction cup **1221**, avoiding delays in assembly progress due to suction cup **1221** damage. Moreover, since the first bracket **121** is provided with multiple groups of first fixing structures, the suction cup base can be detached from one first fixing structure and installed on another first fixing structure, thereby changing an adsorption position of the suction cup **1221**, enabling the suction cup **1221** to avoid harness regions of various models of harness

isolation plates **10**, enhancing compatibility of the suction tool **12**, and achieving rapid type switching.

[0111] In some embodiments, the first fixing structure are multiple groups of suction cup base mounting holes **1222** arranged at different positions on the first bracket **121**. Based on different models of harness isolation plates **10**, multiple groups of suction cup base mounting holes **1222** are provided on the first bracket **121**, each group of suction cup base mounting holes **1222** being arranged on the first bracket **121** to avoid the harness region of a different model of harness isolation plate **10**. The suction cup base is provided with an insertion piece, the insertion piece of the suction cup base can be inserted into the suction cup base mounting hole **1222** so as to achieving connection between the suction assembly **122** and the first bracket **121**.

[0112] Since the first fixing structure includes the suction cup base mounting holes **1222**, the suction cup base cooperates with the suction cup base mounting holes **1222** so as to achieve detachment and installation, allowing convenient and rapid insertion and extraction, and enabling quick adjustment of a position of the suction cup **1221**.

[0113] In some embodiments, the suction cup base is movably connected to the first bracket **121**, so that the suction cup base is movable relative to the first bracket **121** to at least two different positions. The suction cup base is movably connected to the first bracket **121**, and the suction cup base moves relative to the first bracket **121**, allowing the suction cup bases to move closer to or farther from each other, thereby changing a distance between the suction cups **1221**, and enabling the suction cups **1221** to avoid the harness region of the harness isolation plate **10**.

[0114] Since the suction cup base is movably connected to the first bracket **121**, the suction cup base is provided with the suction cup **1221**, and the suction cup base moves relative to the first bracket **121**, thereby changing the position of the suction cup **1221**. Under the condition that a model of the harness isolation plate **10** to be adsorbed changes, the suction cup base can be moved to enable the suction cup **1221** to avoid harness regions of different types of harness isolation plates **10**, enhancing the compatibility of the suction tool **12**.

[0115] In some embodiments, the suction tool **12** further includes a movable mechanism, the movable mechanism being connected to the suction cup base and configured to drive the suction cup base to be movable relative to the first bracket **121**. The suction cup base is movable relative to the first bracket **121** via the movable mechanism, and the suction cup base is provided with the suction cup **1221**, thereby driving the suction cup **1221** to move relative to the first bracket **121**. Under the condition that the model of the harness isolation plate **10** to be adsorbed changes, the suction cup base can be moved to enable the suction cup **1221** to avoid harness regions of different types of harness isolation plates **10**, enhancing the compatibility of the suction tool **12**.

[0116] In some embodiments, the movable mechanism includes a slide rail arranged on the first bracket **121** and a cylinder capable of driving the suction cup base to move, the suction cup base being slidably connected to the slide rail. The slide rail is provided on a surface of the first bracket **121** away from the first connection structure **123**. The slide rail is arranged in a one-to-one correspondence with the suction cup base. The slide rails are spaced apart from each other on the first bracket **121**, and an extension direction of the slide rail is perpendicular to an extension direction of the suction cup base. Each slide rail is slidably connected to one suction cup base, and the suction cup base is movable along the extension direction of the slide rail to drive the suction cup **1221** to move toward or away from each other, so that the suction cup **1221** can avoid the harness region of the harness isolation plate **10**.

[0117] The cylinder may be a single-acting cylinder, a double-acting cylinder, a diaphragm cylinder, or an impact cylinder. This disclosure imposes no specific limitations on a type of the cylinder. A cylinder barrel of the cylinder is arranged on the first bracket **121**, and a piston of the cylinder is connected to the suction cup base to drive the suction cup base to move on the slide rail.

[0118] By arranging the slide rail on the first bracket **121**, the suction cup base is slidably connected to the slide rail, achieving movement of the suction cup base relative to the first bracket

121. Simply pushing the suction cup base can change the position of the suction cup **1221**, making the suction tool compatible with different types of harness isolation plates **10**. The operation is simple, type switching is fast, and time and effort are saved. By providing the cylinder capable of driving the suction cup base, electrical operation can be achieved, offering convenience and avoiding errors from manual operation.

[0119] In some embodiments, the suction tool **12** further includes a first positioning structure arranged on the first bracket **121**, the first positioning structure having a positioning portion arranged along a suction direction of the suction assembly **122**, and the first positioning structure being configured to match a corresponding structure of another tool to prevent relative displacement between the suction tool and the another tool along a non-suction direction. The first bracket **121** is configured in a plate shape, the suction assembly **122** is arranged at a position near a center of the extension surface of the first bracket **121**, the first positioning structure is arranged on the extension surface of the first bracket **121** located between an edge of the suction assembly **122** and an edge of the first bracket **121**, and the suction direction of the suction assembly **122** is perpendicular to the extension surface of the first bracket **121**. The first positioning structure has the positioning portion arranged along the suction direction and is configured to match the corresponding structure of the another tool. In other words, the positioning portion of the first positioning structure is perpendicular to the extension surface of the first bracket **121** and protrudes beyond a plane of the suction assembly **122** away from the first bracket **121**.

[0120] By arranging the first positioning structure on the first bracket **121**, and since the first positioning structure can match the corresponding structure of the other tool, positioning between the suction tool **12** and the other tool is achieved. Since the first positioning structure has the positioning portion arranged along the suction direction of the suction assembly **122**, the suction tool **12** and the another tool cannot displace along the non-suction direction. The structure is simple, and assembly precision is high.

[0121] In some embodiments, the first positioning structure includes a positioning pin **124** oriented toward the suction direction of the suction assembly **122**, the positioning pin **124** being configured to cooperate with a corresponding structure of an assembly platform **16** to prevent relative displacement between the suction tool **12** and the assembly platform **16** along the non-suction direction. The first positioning structure includes the positioning pin **124** arranged perpendicular to the extension surface of the first bracket **121**, and the positioning pin **124** is inserted into the corresponding structure of the assembly platform **16**, such as a positioning hole. A relative position between the positioning pin **124** and the suction assembly **122** on the first positioning structure is fixed, and after the positioning pin **124** is inserted into the corresponding structure of the assembly platform **16**, a relative position of the suction assembly **122** on the assembly platform **16** can also be determined. Multiple positioning pins **124** may be provided, and multiple corresponding structures are provided on the assembly platform **16** accordingly. Insertion is possible only after all positions of the positioning pins **124** align with positions of the corresponding structures of the assembly platform **16**, achieving precise positioning.

[0122] Since the positioning pin **124** cooperates with the corresponding structure of the assembly platform **16**, and the positioning pin **124** is oriented toward the suction direction of the suction assembly **122**, the suction tool **12** and the assembly platform **16** can be positioned via the positioning pin **124**, preventing displacement between the suction tool **12** and the assembly platform **16** along the non-suction direction.

[0123] In some embodiments, the first positioning structure includes a positioning groove **130** arranged on the first bracket **121**, the positioning groove **130** being configured to cooperate with a corresponding structure of a positioning platform to prevent relative displacement between the suction tool **12** and the positioning platform along the non-suction direction. The positioning groove **130** is provided on a surface of the first bracket **121** near the suction assembly **122**, the positioning platform is provided with the corresponding structure cooperating with the positioning

groove **130**, and when the first bracket **121** drives the suction assembly **122** to approach the positioning platform, the corresponding structure on the positioning platform is inserted into the positioning groove **130** and becomes in contact with a bottom of the positioning groove **130**, preventing further approach of the first bracket **121**. The positioning groove **130** extends into the first bracket **121** in a direction perpendicular to the extension surface of the first bracket **121**, meaning that the positioning groove **130** extends into the first bracket **121** parallel to the suction direction. After the corresponding structure on the positioning platform is inserted into the positioning groove **130**, movement between the positioning platform and the suction tool **12** can only occur in the suction direction and not in the non-suction direction. Optionally, multiple positioning grooves **130** are provided, and multiple corresponding structures are provided on the positioning platform accordingly. Insertion is possible only after all positions of the positioning grooves **130** align with positions of the corresponding structures of the positioning platform, achieving precise positioning.

[0124] Since the first bracket **121** is provided with the positioning groove **130**, the positioning groove **130** can cooperate with the corresponding structure of the positioning platform, and after the corresponding structure of the positioning platform cooperates with the positioning groove **130**, displacement between the suction tool **12** and the positioning platform along the non-suction direction is prevented.

[0125] In some embodiments, the suction tool **12** further includes a buffer member **125**, the buffer member **125** being arranged on a side of the suction assembly **122** facing the harness isolation plate **10**, and under the condition that the suction assembly **122** adsorbs the harness isolation plate **10**, the buffer member **125** abutting against the harness isolation plate **10**. The buffer member **125** is provided on a surface of the suction assembly **122** away from the first bracket **121**, and the buffer member **125** itself can deform under external force, thereby counteracting an impact of the external force. The buffer member **125** may be made of an elastic material or an elastic structure. The elastic material includes but is not limited to sponge, foamed polyethylene, and polyester foam. The elastic structure includes but is not limited to a spring.

[0126] The buffer member **125** fully or partially covers a surface of the suction assembly **122**, and the suction assembly **122** is in contact with the harness isolation plate **10** via the buffer member **125**. This disclosure imposes no specific limitations on a specific arrangement form of the buffer member **125**. When the buffer member **125** is arranged to fully cover the surface of the suction assembly **122**, since the suction assembly **122** avoids the harness region of the harness isolation plate **10**, the buffer member **125** arranged thereon also avoids the harness region of the harness isolation plate **10**. When the buffer member **125** is arranged to partially cover the surface of the suction assembly **122**, the buffer member **125** may be arranged corresponding to the palladium sheets **102** with higher strength on the harness isolation plate **10**, so that when the suction assembly **122** adsorbs the harness isolation plate **10**, only the palladium sheets **102** with higher strength become in contact and bear force with the buffer member **125**.

[0127] Since the buffer member **125** is provided on the side of the suction assembly **122** facing the harness isolation plate **10**, when the suction assembly **122** adsorbs the harness isolation plate **10**, the harness isolation plate **10** directly becomes in contact with the buffer member **125** rather than the suction assembly **122**, and the buffer member **125** disperses and cushions pressure between the harness isolation plate **10** and the suction assembly **122**.

[0128] In some embodiments, the buffer member **125** is an elastic pad arranged on the suction assembly **122**, the elastic pad is provided with multiple suction holes **1251**, and the suction assembly **122** adsorbs the harness isolation plate **10** through the suction holes **1251**. The elastic pad is arranged on a surface of the suction assembly **122** away from the first bracket **121** and directly becomes in contact with the harness isolation plate **10** during adsorption. The elastic pad is designed with clearance. Similarly, the elastic pad is arranged to avoid the harness region of the harness isolation plate **10**, and the elastic pad is provided with multiple suction holes **1251** to allow

airflow within the suction assembly **122**, enabling the suction assembly **122** to adsorb the harness isolation plate **10** through the suction holes **1251**. Further, the elastic pad is arranged on the suction assembly **122** corresponding to the palladium sheets **102** of the harness isolation plate **10**, and becomes in contact with only the palladium sheets **102** with higher strength during adsorption, achieving adsorption of the harness isolation plate **10** by adsorbing only the palladium sheets **102**. Optionally, the elastic pad is a sponge pad.

[0129] Further, a pressure sensor may be arranged on the elastic pad to detect a pressure exerted by the elastic pad on the harness isolation plate **10** during adsorption and downward overshoot. If an adsorption force is too large or too small, a vacuum level inside the suction assembly **122** may be appropriately adjusted to regulate the suction force provided by the suction assembly **122**; and if the pressure is too large, a pressure exerted by an external device on the first bracket **121** may be appropriately reduced.

[0130] Since the elastic pad is provided on the side of the suction assembly **122** facing the harness isolation plate **10**, during adsorption, the elastic pad directly becomes in contact with the harness isolation plate **10**, dispersing and cushioning the pressure between the harness isolation plate **10** and the suction assembly **122** through the elasticity of the elastic pad. The elastic pad is provided with the suction holes **1251**, enabling the suction assembly **122** to perform normal adsorption through the suction holes **1251**.

[0131] In some embodiments, the suction tool **12** further includes a vacuum generator **126** assembly arranged on the first bracket **121**, the vacuum generator **126** assembly being in communication with the suction assembly **122** and configured to provide power to the suction assembly **122** for adsorbing the harness isolation plate **10**. By arranging the vacuum generator **126** assembly on the first bracket **121**, power required for the suction assembly **122** to adsorb the harness isolation plate **10** is provided.

[0132] In some embodiments, the vacuum generator **126** assembly includes a vacuum generator **126**, a pneumatic switch valve **127**, and a pneumatic duplex component **128**, the vacuum generator **126** being sequentially connected to the suction assembly **122** via the pneumatic duplex component **128** and the pneumatic switch valve **127**. The vacuum generator **126** and the pneumatic duplex component **128** are both arranged on the first bracket **121**, the vacuum generator **126** is in communication with the suction assembly **122** via the pneumatic duplex component **128**, the vacuum generator **126** is configured to provide power for gas flow, and the pneumatic duplex component **128** is in communication with the suction assembly **122** to filter gas from the suction assembly **122**. Airflow is drawn in from the suction assembly **122**, filtered by the pneumatic duplex component **128**, and then discharged through the vacuum generator **126**. The pneumatic switch valve **127** is arranged between the pneumatic duplex component **128** and the vacuum generator **126**, and changing an opening degree of the pneumatic switch valve **127** can control an amount of gas entering, thereby adjusting the adsorption force. Optionally, the vacuum generator **126** provides a working air pressure of 0.5 MPa and has an air consumption of 180 L/min.

[0133] Specifically, the pneumatic duplex component **128** is connected to the vacuum generator **126** via a gas delivery pipe, and the pneumatic switch valve **127** is arranged on the gas delivery pipe. The pneumatic duplex component **128** is also electrically connected to the pneumatic switch valve **127**, and the pneumatic duplex component **128** can monitor air pressure within the suction assembly **122** and control the opening degree of the pneumatic switch valve **127** based on the air pressure. Optionally, the pneumatic duplex component **128** is electrically connected to the vacuum generator **126** to control opening and closing of the vacuum generator **126** based on the air pressure within the suction assembly **122**. A vacuum level within the suction assembly **122** can be controlled at -50 kPa, which meets suction force requirements for most models of harness isolation plates **10**. For example: when a model of the harness isolation plate **10** is BLX-SW-910x180-10, a suction force provided by the vacuum generator **126** is 560 N; and when the model of the harness isolation plate **10** is BLX-SW-720x180-10, the suction force provided by the vacuum generator **126** is 440

N.

[0134] Power for the suction assembly **122** to adsorb the harness isolation plate **10** is provided through the vacuum generator **126**. The pneumatic duplex component **128** controls an opening degree of the pneumatic switch valve **127** based on the air pressure within the suction assembly **122**, thereby stabilizing the air pressure within the suction assembly **122**. The pneumatic duplex component **128** is further configured to filter air, filtering air from the suction assembly **122**.

[0135] In some embodiments, a safety hole **129** is provided on the first bracket **121** at a position corresponding to the suction assembly **122**, and the safety hole **129** is in communication with the suction assembly **122**. Multiple safety holes **129** are provided through the first bracket **121**, the suction assembly **122** is in communication with the external atmosphere via the safety holes **129**, and under the condition that an internal air pressure of the suction assembly **122** is too low, air can be introduced through the safety holes **129** to maintain internal air pressure stability. By providing the safety hole **129** in communication with the suction assembly **122** on the first bracket **121**, air circulation is maintained, preventing excessively low air pressure inside the suction assembly **122**. Optionally, a pressure relief valve is arranged on the first bracket **121**, the pressure relief valve is in communication with the suction assembly **122**, and the pressure relief valve can automatically open or close based on the internal air pressure of the suction assembly **122** to maintain the air pressure within the suction assembly **122** within a safe range.

[0136] In some embodiments, the first connection structure **123** is a first lifting ring **1231** arranged on the first bracket **121**. The first connection structure **123** may alternatively be a hook, a magnetic piece, or a handle **1232**, provided that the first connection structure can cooperate with the transport device to achieve hoisting. This disclosure imposes no specific limitations on a specific arrangement form of the first connection structure **123**. For example, the first connection structure **123** is a handle **1232** arranged on the first bracket **121**, two handles **1232** are spaced apart from each other on two sides of the first bracket **121**, and the transport device or an operator can move the first bracket **121** by grasping the handles **1232**.

[0137] Optionally, one first lifting ring **1231** is arranged at the center of a surface of the first bracket **121** away from the suction assembly **122**, and the transport device is connected to the first lifting ring **1231** to drive the suction tool **12** to move. Similarly, optionally, two first lifting rings **1231** are arranged spaced apart on the first bracket **121**, and the first lifting rings **1231** are symmetrically arranged on the first bracket **121**. When the transport device drives the first lifting rings **1231** to hoist the suction tool **12**, a posture of the suction tool **12** can be maintained stable. Connection of the first lifting ring **1231** and the transport device allows the suction tool **12** to be hoisted to a designated position.

[0138] In some embodiments, as shown in FIG. 5 and FIG. 6, the harness isolation plate assembly mechanism further includes a positioning tool **14** arranged on a positioning platform, the positioning tool **14** includes a second bracket **141** and a second positioning structure **142**, the second bracket **141** is configured to support the harness isolation plate **10**; and the second positioning structure **142** is arranged on the second bracket **141** and configured to cooperate with the first positioning structure to prevent relative displacement between the suction tool **12** and the positioning platform along the non-suction direction. The second bracket **141** is configured in a plate shape, and a dimension of the second bracket **141** is larger than a dimension of the harness isolation plate **10**. One surface of the second bracket **141** rests on the positioning platform, and the second positioning structure **142** is provided on the other surface of the second bracket **141**. The suction tool **12** moves with the harness isolation plate **10** to above the positioning platform, and through cooperation between the first positioning structure and the second positioning structure **142**, the harness isolation plate **10** of the suction tool **12** is positioned within an upper region of the positioning tool **14**, and the harness isolation plate **10** falls into the positioning tool **14** after being released.

[0139] Since the positioning tool **14** includes the second positioning structure **142**, and the second

positioning structure **142** can cooperate with the first positioning structure, enabling mutual positioning between the positioning tool **14** and the suction tool **12**. Since the first positioning structure has the positioning portion arranged along the suction direction of the suction assembly **122**, relative displacement between the suction tool **12** and the positioning platform along the non-suction direction can be prevented.

[0140] In some embodiments, as shown in FIG. 7, the second positioning structure **142** is a top post arranged on the second bracket **141**, the top post protrudes upward along the second bracket **141**, and under the condition that the top post cooperates with the first positioning structure, the top post supports and lifts the suction tool **12**, so that the suction assembly **122** and an upper surface of the second bracket **141** have a clearance gap. The second positioning structure **142** may alternatively be a groove arranged on the second bracket **141**. The groove recesses into the second bracket **141** to a certain depth, and when the groove cooperates with the first positioning structure, the first positioning structure supports and lifts the suction tool **12** away from the positioning tool **14**, so that the suction assembly **122** and the upper surface of the second bracket **141** have a clearance gap. The surface of the second bracket **141** away from the positioning platform is the upper surface.

[0141] Optionally, the first positioning structure includes a positioning pin **124** oriented toward the suction direction, and after the positioning pin **124** aligns with the top post, the first bracket **121** cannot move closer to the second bracket **141**, creating a gap between the first bracket **121** and the second bracket **141**, where a length of the gap is a sum of a length of the positioning pin **124** and a length of the top post. Alternatively, a length of the positioning pin **124** may be greater than a depth of the groove on the second bracket **141**, and the positioning pin **124** is inserted into a bottom of the groove, creating a gap between the first bracket **121** and the second bracket **141**, where a length of the gap is a difference between the length of the positioning pin **124** and the depth of the groove. Similarly, optionally, the first positioning structure includes a positioning groove **130**, a length of the top post is greater than a depth of the positioning groove **130**, and the top post is inserted into a bottom of the positioning groove **130**, creating a gap between the first bracket **121** and the second bracket **141**, where a length of the gap is a difference between the length of the top post and the depth of the positioning groove **130**. Optionally, the length of the gap is determined based on a production tolerance of the harness isolation plate **10** and ranges from 0.5 mm to 2.0 mm.

[0142] Since the second bracket **141** is provided with the top post protruding upward, after the top post cooperates with the first positioning structure, displacement between the positioning tool **14** and the suction tool **12** in the non-suction direction can be prevented, and the suction tool **12** can be lifted by the top post, maintaining a safety gap between the positioning tool **14** and the suction tool **12**.

[0143] In some embodiments, the positioning tool **14** further includes a limiting member **143** arranged on the second bracket **141**, the limiting member **143** being configured to enclose a positioning space matching an outer contour of the harness isolation plate **10**. The limiting members **143** together form a continuous or discontinuous frame, the frame corresponds to the outer contour of the harness isolation plate **10** to form the positioning space, and under the condition that the harness isolation plate **10** is placed in the positioning space, edges of the harness isolation plate **10** fit closely with the limiting members **143**. During use, the suction tool **12** drives the harness isolation plate **10** to move close to the positioning tool **14**, and after the positions of the harness isolation plate **10** and the positioning tool **14** are determined through the first positioning structure and the second positioning structure **142**, the harness isolation plate **10** is released, the harness isolation plate **10** falls into the positioning space formed by the limiting members **143**, and the limiting members **143** perform position correction, achieving precise positioning of the harness isolation plate **10**.

[0144] If the limiting members **143** are configured as a continuous frame, that is, the limiting members **143** are configured into a closed shape, edges of the frame fit closely with edges of the harness isolation plate **10**. Further, the second bracket **141** is provided with multiple groups of

slots, each group of slots being arranged corresponding to different types of harness isolation plates **10**, meaning that the slots are arranged around a periphery of the harness isolation plate **10**. One side of the limiting member **143** is provided with an insertion piece, and the limiting member **143** is mounted on the second bracket **141** by inserting the insertion piece into the slot. The limiting members **143** are configured in various forms based on different models of harness isolation plates **10**, with edges of each type of limiting member **143** respectively fitting a contour of the corresponding type of harness isolation plate **10**. During use, a type of the limiting member **143** is determined based on a type of the harness isolation plate **10**, and the corresponding type of limiting member **143** is inserted into the slot, enabling the positioning tool **14** to adapt to different types of harness isolation plates **10**.

[0145] If the limiting members **143** are configured as a discontinuous frame, that is, the limiting members **143** are configured into an open shape, edges of the frame fit closely with edges of the harness isolation plate **10**. At least four limiting members **143** are provided, the limiting members **143** are arranged in pairs opposite each other and spaced apart on the second bracket **141**. The limiting members **143** may be configured in an L-shape and arranged on the second bracket **141** corresponding to four corners of the harness isolation plate **10**; alternatively, the limiting members **143** may be configured as flat plates and arranged on the second bracket **141** corresponding to four edges of the harness isolation plate **10**. The limiting members **143** together enclose the positioning space for supporting and positioning the harness isolation plate **10**.

[0146] By arranging the limiting member **143** on the second bracket **141**, and since the limiting member **143** encloses the positioning space matching the outer contour of the harness isolation plate **10**, after the suction tool **12** moves the harness isolation plate **10** to the positioning tool **14** and positioning is achieved through cooperation between the first positioning structure and the second positioning structure **142**, the harness isolation plate **10** is released so that the harness isolation plate **10** to fall into the positioning space enclosed by the limiting member **143**, achieving position correction of the harness isolation plate **10**.

[0147] In some embodiments, a side of the limiting member **143** facing the positioning space is provided with a guide slope, and an upper end of the guide slope is inclined toward a direction away from the positioning space, so that a planar contour of an end of the positioning space away from the second bracket **141** is larger than a planar contour of an end of the positioning space close to the second bracket **141**. The limiting member **143** is arranged on a surface of the second bracket **141** away from the positioning platform and protrudes from the surface of the second bracket **141**. The limiting members **143** together enclose the positioning space, and a surface of the limiting member **143** facing the positioning space is configured as the guide slope with a narrow upper end and a wide lower end.

[0148] Specifically, the upper end of the guide slope is inclined toward the direction away from the positioning space, one side of the space is provided with the guide slope, and the upper end of the guide slope is inclined toward the direction away from the positioning space, so that the planar contour of the end of the positioning space away from the second bracket **141** is larger than the planar contour of the end of the positioning space close to the second bracket **141**. An end of the guide slope of the limiting member **143** close to the second bracket **141** is considered a lower end, and an end away from the second bracket **141** is considered an upper end.

[0149] Since the guide slope is provided on the side of the limiting member **143** facing the positioning space, and the upper end of the guide slope is inclined toward the direction away from the positioning space, even if the falling harness isolation plate **10** has a certain deviation relative to the positioning space, the harness isolation plate **10** can be easily guided into the positioning space.

[0150] In some embodiments, the limiting member **143** is slidably connected to the second bracket **141**, allowing the limiting member **143** to slide and adjust a size of the positioning space. Multiple limiting members **143** are provided, at least one limiting member **143** corresponds to one edge of the harness isolation plate **10**, and each limiting member **143** is slidably connected to the second

bracket **141**, enabling opposing limiting members **143** to slide toward or away from each other, so that the positioning space enclosed by the limiting members **143** can adapt to different models of harness isolation plates **10**. An oblong hole **144** may be provided on the second bracket **141**, an extension direction of the oblong hole **144** is perpendicular to an extension direction of the limiting member **143**, and the limiting member **143** is inserted into the oblong hole **144** and can slide along the oblong hole **144**. Sliding the limiting member **143** changes a distance between opposing limiting members **143**, enabling the limiting members **143** to fit edges of different models of harness isolation plates **10**.

[0151] By slidably connecting the limiting member **143** to the second bracket **141**, the limiting member **143** can slide relative to the second bracket **141** to change a position of the limiting member **143** relative to the second bracket **141**, thereby changing a distance between the limiting members **143** and adjusting the size of the positioning space. This allows the limiting member **143** to be suitable for positioning harness isolation plates **10** of different sizes, offering strong compatibility.

[0152] In some embodiments, the second bracket **141** is provided with multiple support members **145**, the multiple support members **145** are arranged to avoid the harness region of the harness isolation plate **10** to support the harness isolation plate **10**. Multiple support members **145** such as support posts are arranged on a surface of the second bracket **141** away from the positioning platform. The multiple support members **145** are arranged on the second bracket **141** avoiding the harness region. The support members **145** may be arranged on the second bracket **141** corresponding to the palladium sheets **102** of the harness isolation plate **10**. Further, an end of the support member **145** away from the second bracket **141** is provided with a cushion to buffer an impact force when the harness isolation plate **10** becomes in contact with the support member **145** after being released.

[0153] Optionally, multiple support members **145** are arranged within the positioning space, the support members **145** are arranged avoiding the harness region, and a height of the support member **145** is lower than a height of the limiting member **143**. During use, the harness isolation plate **10** falls into the positioning space, edges of the harness isolation plate **10** fit closely with edges of the limiting members **143**, and the support members **145** become in contact with the palladium sheets **102** and the plastic plate **103** of the harness isolation plate **10** to provide support.

[0154] By arranging the support members **145** avoiding the harness region on the second bracket **141**, after the harness isolation plate **10** is released and falls onto the positioning tool **14**, the harness region with the lowest strength on the harness isolation plate **10** is suspended, and only non-harness regions are in contact with the support members **145**. This prevents impact force from damaging the harness region when the harness isolation plate **10** falls onto the positioning tool **14**.

[0155] In some embodiments, the second bracket **141** is further provided with a second connection structure **146**, the second connection structure **146** being configured to connect to the transport device to move the positioning tool **14**. The second connection structure **146** may be configured as a lifting ring, a hook, or a magnetic piece, provided that the second connection structure can cooperate with the transport device to achieve hoisting. This disclosure imposes no specific limitations on the second connection structure **146**. The second connection structure **146** is arranged on a surface of the second bracket **141** away from the positioning platform, and the transport device is connected to the second connection structure **146** to drive the suction tool **12** to move.

[0156] In some embodiments, the positioning tool **14** further includes support legs **147**, the support legs **147** being connected to the second bracket **141** to support the second bracket **141** and adjust a height of the second bracket **141**. Specifically, four support legs **147** are provided and arranged near four corners of the second bracket **141**, respectively. A surface of the support leg **147** is configured with threads, and the support leg **147** is connected to the second bracket **141** via threads. After the second bracket **141** is placed on the positioning platform, the support legs **147** can be

rotated to adjust a distance of the second bracket **141** relative to the positioning platform or to level the second bracket **141**.

[0157] By arranging the support legs **147** on the second bracket **141**, support for the positioning tool **14** is achieved. Additionally, the support legs **147** can be further configured to adjust the height of the second bracket **141**, facilitating height adaptation between the positioning tool **14** and the suction tool **12**.

[0158] In some embodiments, the harness isolation plate assembly mechanism further includes a loading tray, the loading tray being arranged on a loading platform and configured to support the harness isolation plate **10** to be adsorbed; and the loading tray being provided with a third positioning structure configured to cooperate with the first positioning structure to prevent relative displacement between the suction tool **12** and the loading platform along the non-suction direction. The loading tray is placed on the loading platform, one surface of the loading tray is in contact with the loading platform, and the other surface supports the harness isolation plate **10**. The third positioning structure is provided on the side supporting the harness isolation plate **10**. The third positioning structure can cooperate with the first positioning structure, enabling mutual positioning between the loading tray and the suction tool **12**.

[0159] Since the loading tray is provided with the third positioning structure cooperating with the first positioning structure, and the first positioning structure is arranged on the suction tool **12**, after the first positioning structure cooperates with the third positioning structure, a position between the suction tool **12** and the loading tray is fixed. Since the first positioning structure has the positioning portion arranged along the suction direction of the suction assembly **122**, relative displacement between the suction tool **12** and the loading platform along the non-suction direction can be prevented.

[0160] According to a second aspect, this disclosure provides a battery production line, as shown in FIG. **8**, including: an assembly platform **16**; the harness isolation plate **10** assembly mechanism as described above; and a transport device, the transport device being at least configured to transport the harness isolation plate **10** assembly mechanism to the assembly platform **16**.

[0161] Since the battery production line is equipped with the harness isolation plate **10** assembly mechanism described above, during adsorption of the harness isolation plate **10**, low-strength regions on the harness isolation plate **10** can be avoided, and the force on the harness isolation plate **10** is dispersed, achieving damage-free assembly of the harness isolation plate **10**.

[0162] In some embodiments, the harness isolation plate **10** assembly mechanism includes a suction tool **12**, a positioning tool **14**, and a loading tray; and the battery production line further includes a loading platform and a positioning platform, the loading tray being arranged on the loading platform, the positioning tool **14** being arranged on the positioning platform; and the transport device being at least configured to transport the suction tool **12** from the loading platform to the positioning platform and from the positioning platform to the assembly platform **16**.

[0163] The suction tool **12** adsorbs the harness isolation plate **10** from the loading tray on the loading platform, the transport device moves the suction tool **12** to the positioning platform, the suction tool **12** releases the harness isolation plate **10** to the positioning tool **14** on the positioning platform for positioning, and the suction tool **12** adsorbs the positioned harness isolation plate **10** to the assembly platform **16** to complete subsequent assembly. This achieves precise assembly.

[0164] In some embodiments, the suction tool **12** is provided with a first positioning structure, the positioning tool **14** is provided with a second positioning structure **142**, the loading tray is provided with a third positioning structure, and the assembly platform **16** is provided with a fourth positioning structure **161**; where the first positioning structure is capable of cooperating with the second positioning structure **142** to restrict relative displacement between the suction tool **12** and the positioning tool **14**; the first positioning structure is capable of cooperating with the third positioning structure to restrict relative displacement between the suction tool **12** and the loading tray; and the first positioning structure is capable of cooperating with the fourth positioning

structure **161** to restrict relative displacement between the suction tool **12** and the assembly platform **16**.

[0165] Positioning between the suction tool **12** and the positioning tool **14** can be achieved through cooperation between the first positioning structure and the second positioning structure **142**; positioning between the suction tool **12** and the loading tray can be achieved through cooperation between the first positioning structure and the third positioning structure; and positioning between the suction tool **12** and the assembly platform **16** can be achieved through cooperation between the first positioning structure and the fourth positioning structure **161**. This achieves precise assembly.

[0166] The foregoing embodiments are for description of the technical solutions of this disclosure only rather than for limiting this disclosure. Although this disclosure has been described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should appreciate that they can still make modifications to the technical solutions described in the embodiments or make equivalent replacements to some or all technical features thereof. However, these modifications or replacements do not make the essence of the corresponding technical solutions deviate from the scope of the technical solutions of the embodiments of this disclosure, and they should all be included in the scope of this disclosure. In particular, as long as there is no structural conflict, the various technical features mentioned in the embodiments can be combined in any manner.

Claims

1. A harness isolation plate assembly mechanism comprising: a suction tool comprising a first bracket and a suction assembly; wherein the first bracket is provided with a first connection structure, the first connection structure being configured to connect to a transport device to move the first bracket, and an extension surface of the first bracket being parallel to an extension surface of a harness isolation plate to be grabbed; and the suction assembly is fixed on the extension surface of the first bracket and arranged to avoid a harness region of the harness isolation plate, and the suction assembly is configured to adsorb or release the harness isolation plate.
2. The harness isolation plate assembly mechanism according to claim 1, wherein the suction assembly is arranged at a position on the first bracket away from an edge and corresponds to a palladium sheet of the harness isolation plate.
3. The harness isolation plate assembly mechanism according to claim 1, wherein the suction assembly comprises a suction cup and a suction cup base, the suction cup base being connected to the first bracket, and the suction cup being arranged on the suction cup base and configured to adsorb or release the harness isolation plate.
4. The harness isolation plate assembly mechanism according to claim 3, wherein the first bracket is provided with multiple groups of first fixing structures, the suction cup base is detachably connected to the first bracket via any one group of the first fixing structures.
5. The harness isolation plate assembly mechanism according to claim 3, wherein the suction cup base is movably connected to the first bracket, so that the suction cup base is movable relative to the first bracket to at least two different positions.
6. The harness isolation plate assembly mechanism according to claim 5, wherein the suction tool further comprises a movable mechanism, wherein the movable mechanism is connected to the suction cup base and configured to drive the suction cup base to be movable relative to the first bracket.
7. The harness isolation plate assembly mechanism according to claim 6, wherein the movable mechanism comprises a slide rail arranged on the first bracket and a cylinder capable of driving the suction cup base to move, the suction cup base being slidably connected to the slide rail.
8. The harness isolation plate assembly mechanism according to claim 1, wherein the suction tool further comprises a first positioning structure arranged on the first bracket, the first positioning

structure having a positioning portion arranged along a suction direction of the suction assembly, and the first positioning structure being configured to match a corresponding structure of another tool to prevent relative displacement between the suction tool and the another tool along a non-suction direction.

9. The harness isolation plate assembly mechanism according to claim 1, wherein the suction tool further comprises a buffer member, the buffer member being arranged on a side of the suction assembly facing the harness isolation plate, and under the condition that the suction assembly adsorbs the harness isolation plate, the buffer member abutting against the harness isolation plate.

10. The harness isolation plate assembly mechanism according to claim 1, wherein the suction tool further comprises a vacuum generator assembly arranged on the first bracket, the vacuum generator assembly being in communication with the suction assembly and configured to provide power to the suction assembly for adsorbing the harness isolation plate.

11. The harness isolation plate assembly mechanism according to claim 10, wherein the vacuum generator assembly comprises a vacuum generator, a pneumatic switch valve, and a pneumatic duplex component, the vacuum generator being sequentially connected to the suction assembly via the pneumatic duplex component and the pneumatic switch valve.

12. The harness isolation plate assembly mechanism according to claim 1, wherein the first connection structure is a first lifting ring arranged on the first bracket.

13. The harness isolation plate assembly mechanism according to claim 8, wherein the harness isolation plate assembly mechanism further comprises a positioning tool arranged on a positioning platform, the positioning tool comprises a second bracket and a second positioning structure, and the second bracket is configured to support the harness isolation plate; and the second positioning structure is arranged on the second bracket and configured to cooperate with the first positioning structure to prevent relative displacement between the suction tool and the positioning platform along the non-suction direction.

14. The harness isolation plate assembly mechanism according to claim 13 wherein the second positioning structure is a top post arranged on the second bracket, the top post protrudes upward along the second bracket, and under the condition that the top post cooperates with the first positioning structure, the top post supports and lifts the suction tool, so that the suction assembly and an upper surface of the second bracket have a clearance gap.

15. The harness isolation plate assembly mechanism according to claim 13, wherein the positioning tool further comprises a limiting member arranged on the second bracket, the limiting member being configured to enclose a positioning space matching an outer contour of the harness isolation plate.

16. The harness isolation plate assembly mechanism according to claim 15, wherein the limiting member is slidably connected to the second bracket, allowing the limiting member to slide and adjust a size of the positioning space.

17. The harness isolation plate assembly mechanism according to claim 13, wherein the second bracket is provided with multiple support members, the multiple support members are arranged to avoid the harness region of the harness isolation plate to support the harness isolation plate.

18. The harness isolation plate assembly mechanism according to claim 13, wherein the second bracket is further provided with a second connection structure, the second connection structure being configured to connect to the transport device to move the positioning tool.

19. A battery production line comprising: an assembly platform; the harness isolation plate assembly mechanism according to claim 1; and a transport device, the transport device being at least configured to transport the harness isolation plate assembly mechanism to the assembly platform.

20. The battery production line according to claim 19, wherein the harness isolation plate assembly mechanism comprises a suction tool, a positioning tool, and a loading tray; and the battery production line further comprises a loading platform and a positioning platform, the loading tray

being arranged on the loading platform, the positioning tool being arranged on the positioning platform; and the transport device being at least configured to transport the suction tool from the loading platform to the positioning platform and from the positioning platform to the assembly platform; wherein the suction tool is provided with a first positioning structure, the positioning tool is provided with a second positioning structure, the loading tray is provided with a third positioning structure, and the assembly platform is provided with a fourth positioning structure; wherein the first positioning structure is capable of cooperating with the second positioning structure to restrict relative displacement between the suction tool and the positioning tool; the first positioning structure is capable of cooperating with the third positioning structure to restrict relative displacement between the suction tool and the loading tray; and the first positioning structure is capable of cooperating with the fourth positioning structure to restrict relative displacement between the suction tool and the assembly platform.
